

Software Engineering: Requirements Engineering & UML Modelling

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Problem Statement :- Digital Gift Card & Loyalty Point Exchange

1. Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
1	Functional	The system shall validate merchant loyalty points, convert them into universal exchange credits at the applicable dynamic rate, and generate a single-use digital voucher code.	High	Pass: Valid points are converted correctly and a unique voucher code is generated. Fail: Invalid, expired, or duplicate points/vouchers are accepted.	This is the core functionality of the platform.
2	Functional	The system shall allow an Account Holder to connect and manage loyalty point balances from supported Merchant Partners.	High	Pass: Connected merchant balances are displayed correctly. Fail: An unsupported or invalid merchant account is connected.	Enables users to aggregate loyalty points from multiple merchants.
3	Functional	The system shall display the current exchange rate and calculate the number of universal exchange credits before the	High	Pass: The displayed credits match the configured exchange rate. Fail: Incorrect credits are calculated or the rate is unavailable.	Provides transparency before users convert their points.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
4	Functional	Account Holder confirms the conversion.	High		
		The system shall allow an Account Holder to redeem available exchange credits for an eligible digital gift card.		Pass: Sufficient credits are deducted and a valid digital gift card/voucher is issued. Fail: Redemption succeeds when the account has insufficient credits.	Allows users to use their converted value for rewards.
5	Functional	The system shall lock a voucher during redemption and prevent duplicate or fraudulent redemption attempts.	High	Pass: A voucher can be redeemed only once and its status changes to Redeemed. Fail: The same voucher is successfully redeemed multiple times.	Prevents fraud and protects the value of digital gift cards.

2. Non-Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
1	Performance & Security	The system shall process point conversion and gift card redemption operations within 250 ms while maintaining zero ledger balance drift.	High	Pass: Performance testing confirms operations complete within 250 ms and ledger balances remain consistent. Fail: Operations exceed the target or balance discrepancies occur.	Ensures fast and financially accurate transactions.
2	Security & Reliability	The system shall securely protect user accounts, loyalty balances, exchange credits, and voucher transactions using authenticated access, encrypted communication, and transaction logging.	High	Pass: Unauthorized users cannot access another user's account or redeem vouchers; security testing confirms protected communication and complete transaction logs. Fail: Unauthorized access or untraceable voucher transactions occur.	Protects financial value and reduces fraud.

3. Core Use-Case Flow Specification

Section	Details
Use Case Name & ID	UC-01: Redeem Loyalty Points for Digital Gift Card
Goal	To enable an Account Holder to redeem eligible loyalty points for a digital gift card. The system validates the available loyalty points, converts them into exchange credits where applicable, generates a unique digital voucher code, securely records the transaction, and provides the Account Holder with a usable digital gift card.
Primary Actor	Account Holder / Customer
Secondary Actors	Merchant Loyalty Program, Digital Gift Card Service, Merchant System
Preconditions	1. Account Holder is authenticated and logged into the Digital Gift Card & Loyalty Point Exchange platform. 2. Account Holder has an active loyalty account with available loyalty points. 3. A valid digital gift card option is available for redemption.
Postconditions	1. Required loyalty points are successfully deducted and converted into exchange credits where applicable. 2. A unique digital gift card or voucher code is generated. 3. The redemption transaction is recorded in the Account Holder's transaction history. 4. Confirmation of successful redemption is provided to the Account Holder.
Main Success Scenario (MSS)	1. Account Holder logs into the Digital Gift Card & Loyalty Point Exchange platform. 2. System retrieves and displays the Account Holder's available loyalty point balance (UC-01.1 «include»). 3. Account Holder selects a merchant and an available digital gift card. 4. System displays the required loyalty points and/or exchange credits for the selected gift card.

Section

Details

5. Account Holder confirms the redemption request.
6. System validates that the Account Holder has sufficient eligible loyalty points.
7. System converts the required loyalty points into exchange credits according to the applicable conversion rate.
8. System deducts the required loyalty points or exchange credits from the Account Holder's balance.
9. System generates a unique digital gift card or voucher code.
10. System records the transaction and displays a successful redemption confirmation with the voucher details.

6a. Insufficient Loyalty Points:

1. System detects insufficient loyalty points or exchange credits.
2. System displays an insufficient balance message.
3. No points or credits are deducted.
4. Account Holder may select another gift card or cancel the redemption.

3a. Selected Digital Gift Card Unavailable:

1. Account Holder selects a temporarily unavailable gift card.
2. System informs the Account Holder that the gift card is unavailable.
3. No loyalty points are deducted.
4. Account Holder returns to the available gift card list.

Alternate Flows

9a. Digital Voucher Generation Failure:

1. System is unable to generate the digital voucher code due to a service failure.
2. System prevents the redemption from being completed.
3. No loyalty points or exchange credits are permanently deducted.
4. System displays an error message and allows the Account Holder to try again later.

Special Requirements

The system should securely process loyalty point redemption and generate the digital voucher within an acceptable response time. Each successful redemption must generate a unique voucher code and prevent duplicate redemption attempts.

4. UML Use-Case Diagram

The UML Use-Case Model illustrates interactions between key actors and the **Digital Gift Card & Loyalty Point Exchange System** boundary. It explicitly includes the required «include» and «extend» relationships:

Actors

1. **Account Holder / Customer**
2. **Merchant Loyalty Program**
3. **Digital Gift Card Service**
4. **Merchant System**
5. **Administrator**

Main Use Cases

- Register / Login
- View Loyalty Points
- Select Merchant
- Browse Gift Cards
- Redeem Loyalty Points
- Convert Points to Exchange Credits
- Generate Digital Voucher
- Validate Voucher
- View Transaction History
- Manage Gift Cards
- Manage Merchants

- Manage Users

Required Relationships

«include»

Redeem Loyalty Points

→ «include» **Validate Loyalty Points**

→ «include» **Convert Points to Exchange Credits**

→ «include» **Generate Digital Voucher**

«extend»

Apply Promotional Offer

→ «extend» **Redeem Loyalty Points**

Visual UML Structural Representation:



